Query-Variation Robustness of Sparse, Latent-Semantic and Fused First-Stage Retrieval: A Reproducible Pilot Study on the Vaswani Collection

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Abstract

Query-variation robustness asks whether a retrieval system keeps working when the same information need is worded differently. This paper reports a small pilot on the Vaswani/NPL collection (11,429 abstracts, 93 queries), with the analysis plan written before the run. It compares three first-stage systems: BM25, a non-neural latent-semantic-analysis (LSA) retriever, and their reciprocal-rank-fusion (RRF) hybrid. Using the four meaning-preserving categories of Penha et al., 644 query rewrites were kept for analysis. Misspellings changed nDCG@10 by -18.5% (BM25), -21.4% (LSA) and -20.2% (hybrid); all three Holm-adjusted p-values are below 0.001. Paraphrasing changed BM25 by -13.3% (Holm $p = 0.004$), whereas the changes for LSA (-5.3%) and the hybrid (-7.8%) were not significant after correction. Natural-question rewrites and stop-word removal moved nDCG@10 by less than 2% for every system, and word-order changes had exactly zero effect, as they must for bag-of-words systems. Paraphrasing changed the top ten results more than misspelling did but cost less effectiveness, so ranking stability did not track harm across categories. Within each system it did track harm across individual rewrites (Spearman correlation -0.60 to -0.67). The pilot has hard limits. It uses one small, old collection and no neural retriever. No rewrite was checked by a human: typos are assumed to keep their meaning because of how they are made, and the other rewrites were reviewed by a single LLM from the generator's model family. The literature search was not systematic. Code, rewrites, rankings and 85 automated tests are released.

Keywords: information retrieval; query variation; retrieval robustness; BM25; latent semantic analysis; rank fusion; reproducibility

Status of this draft. Pilot study. The code, the rewrites, the experiments and this text were produced by an AI system (Section 8). No human has yet read, verified or approved it. Neural dense retrievers, BEIR datasets and human validation of the query rewrites are *not* part of it. Text in square brackets is for the human author to complete; only people who made a substantial scholarly contribution, have read and verified this work, and accept responsibility for it may be listed as authors.

1 Introduction

A retrieval benchmark fixes one wording of each information need, but users word the same need in many ways. Whether a system's score survives a change of wording is a different question from how high the score is. Query variability has a long research history: the UQV100 test collection was created to study it (Bailey et al., 2016); later work examines retrieval consistency in the presence of query variations (Bailey et al., 2017) and evaluation pooled over query variations (Moffat et al., 2015); and a line of work shows that neural rankers and retrievers lose effectiveness when queries contain typos (Zhuang and Zuccon, 2021, 2022;

Sidiropoulos and Kanoulas, 2022; Tasawong et al., 2023). Penha et al. (2022) organise query variations into a taxonomy, provide generators for them and show that retrieval pipelines lose a substantial share of their effectiveness under syntax-changing variations; Hagen et al. (2024) reproduce that study and extend it to newer neural models.

This paper does not claim that query-variation robustness is a new problem. It is a small pilot with a narrower aim: to build and check a transparent pipeline, and to measure on one collection how three first-stage systems respond to rewording. The systems are BM25, a non-neural latent-semantic retriever and their reciprocal-rank-fusion hybrid (Cormack et al., 2009). The rewrites come from the four meaning-preserving categories of Penha et al. (2022), and for each system the study looks at effectiveness and at how much the top-ranked results move. The statistics are paired and clustered by query, and every analysis step was fixed in writing before the run.

Two research questions are addressed. **RQ1:** how does effectiveness change under each variation category? **RQ2:** do the three systems respond differently? Two further questions from the original research plan, whether effects vary across domains (RQ3) and whether degradation can be predicted on held-out domains (RQ4), are out of scope because the pilot has one collection.

The work was done in a sandbox that could install software packages but could not download neural model weights or BEIR datasets (Thakur et al., 2021). Everything that could be run was run. The code for the rest was written and tested with a stand-in encoder but has never been run on real data. Nothing in this paper is a result about neural dense retrievers or about other domains.

The contributions are modest and are stated as such:

1. An open, tested pipeline (BM25, LSA, RRF, five effectiveness and stability measures, cluster bootstrap, permutation tests with Holm correction) with an account of the defects found in the starter code this work began from (Section 3.7).
2. Effect sizes with intervals for the four categories and three systems on one collection (Section 4), computed from a plan written before the run (Section 3.6).
3. Three observations that are worth testing at scale rather than believing from this pilot: ranking stability and effectiveness loss are not interchangeable across categories; the "paraphrasing" category hides a large difference between generators; and a hybrid of two whole-word systems lies between its components rather than beating both (Sections 4.3 to 4.5).
4. A clear account of what the validity of the rewrites does and does not rest on (Section 3.3), because that validity is the weakest link in this kind of study.

2 Related work

2.1 Query variability and ranking consistency

UQV100 is a test collection created to study query variability (Bailey et al., 2016). Bailey et al. (2017) study retrieval consistency in the presence of query variations, and Moffat et al. (2015) study evaluation pooled over query variations. More recently, Alaofi et al. (2023) explored whether generative language models can create query variants for test collections, and Campese et al. (2025) propose a training loss that penalises differences between the documents retrieved for semantically equivalent queries (read at abstract level only). This line of work establishes that wording sensitivity and ranking consistency are old topics; the contribution of the present pilot cannot be the question itself.

2.2 Robustness of neural retrievers and typos

A group of papers examines typos in neural retrieval and proposes training remedies: BERT-based passage retrieval and ranking (Zhuang and Zuccon, 2021), dense retrievers with CharacterBERT and self-teaching (Zhuang and Zuccon, 2022), an analysis of dual encoders against misspellings (Sidiropoulos and Kanoulas, 2022), typo-robust representation learning (Tasawong et al., 2023) and multi-positive contrastive learning (Sidiropoulos and Kanoulas, 2024). Penha et al. (2022) define a taxonomy of query variations with generators and evaluate BM25 followed by neural re-rankers on two collections, reporting a loss of roughly one fifth of effectiveness on average under their variations (read through a page summariser; see Appendix A). Hagen et al. (2024) reproduce that study for BERT- and T5-based re-rankers and extend it to Sentence-BERT, CharacterBERT-based, E5-Mistral, AnglE and Ada v2 models; from the parts read, they evaluate the re-ranking stage rather than the whole pipeline. Li et al. (2026) study LLM-based dense retrievers and report that they are more robust to typos than encoder-only baselines but remain vulnerable to semantic perturbations such as synonym substitution (abstract only), and Perçin et al. (2025) report that retrievers in retrieval-augmented generation degrade under minor query variations (abstract only).

2.3 Fusion, and the gap this pilot does and does not address

Reciprocal rank fusion combines rankings by summing reciprocal ranks and is a simple, nearly parameter-free way to merge a lexical and a semantic ranking (Cormack et al., 2009). Whether a fused first stage inherits, averages or compounds the sensitivities of its components under query variation is an empirical question. In the sources read for this study (Appendix A), robustness is measured for re-rankers placed on top of BM25 (Penha et al., 2022; Hagen et al., 2024) or for single dense or LLM-based retrievers (Li et al., 2026; Perçin et al., 2025), and none of them, as far as could be seen from the parts read, reports a fused lexical-semantic first stage next to its own components under the Penha et al. (2022) categories with paired, query-clustered inference.

This observation is weak evidence of a gap, not a demonstration. The literature search was not systematic (Appendix A), several sources were read only at abstract level, and the 2025 and 2026 literature is large. The pilot does not close the gap either: its "semantic" component is non-neural LSA, so it cannot test whether a neural retriever and BM25 complement each other under query variation. The most that can be claimed is a transparent, reproducible starting point.

3 Experimental design

3.1 Collection

The Vaswani (NPL) collection has 11,429 abstracts of physics and electronics literature, 93 queries and 2,083 relevance judgments, all binary; every query has at least one relevant document and no judged document is missing from the corpus. The counts agree with the description on the ir_datasets page (ir_datasets, 2026). The collection is small and old, and neither that page nor the Glasgow IR group page for the collection (University of Glasgow, Information Retrieval Group, 2026) states its provenance details or a licence; the latter also carries a caution about the collection. For that reason the collection is not redistributed: `scripts/fetch_vaswani.py` downloads it from a pinned commit of the PyTerrier repository and verifies SHA-256 and md5 checksums (the md5 values equal those PyTerrier itself lists for its Vaswani resources; no checksum from the collection's original authors was found). Whoever runs the script is responsible for checking the terms of use.

3.2 Systems

All systems return their top 100 documents and share one analyzer: lowercasing, `[a-z0-9]+` tokens, the 33-word Lucene English stop list and Porter stemming (Porter, 1980).

- **BM25** (Robertson and Zaragoza, 2009): an own sparse-matrix implementation with Lucene-style IDF, $k_1 = 1.20$ and $b = 0.75$ (common default values; not tuned on any query), query-term multiplicity counted, documents with score zero not returned. Its scores were checked against the `bm25s` package.
- **LSA**: TF-IDF with sublinear term frequency on the same analyzer, truncated SVD to 300 dimensions (random seed 0) (Deerwester et al., 1990), L2 normalisation and exact cosine search. LSA is a latent-semantic retriever, **not a neural one**. It stands in for the "dense" component of the original plan and is not evidence about neural dense retrievers. Like BM25 it works on whole words of the collection's vocabulary.
- **RRF hybrid**: reciprocal rank fusion with $k = 60$ over the two top-100 lists (Cormack et al., 2009).

A sentence-embedding retriever and its hybrid are implemented and unit-tested with a stand-in encoder but were not run, because no model weights were available.

3.3 Query variations and their validity

Categories follow the four meaning-preserving classes of Penha et al. (2022): misspelling, ordering, naturality and paraphrasing. The two meaning-changing classes (generalisation and specialisation, aspect change) are not studied. Generator names follow theirs where they exist; the code is an independent re-implementation. Table 1 lists the generators and what happened to their 734 candidate rewrites.

Table 1. Query rewrites by generator. *Candidates* are all rewrites produced; *kept* are those used in the analysis. No rewrite is human-verified (Section 3.3).

Category	Generator	Produced by	Candidates	Kept	Basis of validity
misspelling	`NeighbCharSwap`	seeded code	93	93	by construction
misspelling	`RandomCharSub`	seeded code	93	93	by construction
misspelling	`QWERTYCharSub`	seeded code	93	93	by construction
paraphrasing	`LLM_Paraphrase`	AI assistant	93	93	LLM review
paraphrasing	`LLM_SynonymSwap`	AI assistant	93	92	LLM review
naturality	`LLM_NaturalQuestion`	AI assistant	93	92	LLM review
naturality	`RemoveStopWords`	code	83	80	LLM review + assistant recheck
ordering	`RandomOrderSwap`	seeded code	93	8	LLM review + function-word rule
all			**734**	**644**	

Misspellings are seeded character edits of one word of at least four letters (adjacent swap, random substitution, substitution by a neighbouring key on a QWERTY keyboard). They are treated as meaning-preserving by construction. That is an assumption: an edit that happens to create another real word could change the meaning. Section 4.7 counts how often that happened with respect to the collection's vocabulary (0 of 279). *Word-order swaps* exchange two words with different surface forms, and *stop-word removal* deletes every word of the NLTK English stop-word list. Both are machine-made (455 machine-made candidates in all, including the typos).

The remaining 279 rewrites (a natural-language question, a differently structured paraphrase and a synonym swap for each query) were written by hand in three batches by the AI assistant that also wrote the code. There

was no separate prompt, sampling setting or seed to record, so the rewrites are stored verbatim. The assistant saw the queries and the collection's vocabulary but no retrieval output.

Validity of the non-typo rewrites. A reviewer saw only the (original, rewrite) pairs, in shuffled order with opaque identifiers and without category or generator, and labelled each pair `equivalent`, `broader`, `narrower` or `different`. The reviewer was three sequential agents run with a smaller Claude model setting. Generator and reviewer belong to the same model family, so shared blind spots are possible. The reviewer prompt is reproduced verbatim in the repository. Thirty typo rewrites were mixed into the packets as a calibration sample, and the reviewer labelled all of them `equivalent`; that result is a diagnostic only and never changed the status of a typo.

The reviewer accepted 47 of the 93 word-order swaps, including garbled ones, which the assistant judged too lenient. Before any retrieval output on the pilot queries had been seen, a stricter second rule was therefore added (amendment A4): a swap is accepted only if every word that moved is a function word, and a rewrite of the swap or stop-word kind enters the analysis only if reviewer and rule agree. The same assistant that wrote the code and the rewrites made the rule and the recheck, so they are stricter but not independent. Of the 734 candidates, 644 were kept (279 by construction and 365 after LLM review); 90 were dropped, 85 of them word-order swaps.

No rewrite in this pilot is human-verified. The analysis is therefore conditional on the rewrites being meaning-preserving, which has been checked only by construction and by one LLM reviewer. A blind sheet of 200 rewrites for a human audit and a tool that scores it are part of the repository, but the audit has not been done.

3.4 Measures

The primary measure is nDCG@10 (linear gain, logarithmic discount) (Järvelin and Kekäläinen, 2002). Secondary measures are Recall@100 and MRR@10 (binary relevance), overlap@10 (the share of the top ten shared by the rankings for the original query and its rewrite) and extrapolated rank-biased overlap (RBO, $p = 0.9$, depth 100) (Webber et al., 2010) between those two rankings. Queries without a relevant document would be excluded; there are none.

3.5 Estimands and inference

For a system and a category, let Q_c be the original queries that have at least one kept rewrite in category c. For each such query the metric of its rewrites is averaged first, so a query counts once per category however many rewrites it has. The primary estimand is the relative change, the mean over Q_c of the rewrite metric divided by the mean over Q_c of the original-query metric, minus one. Intervals are 95% percentile **cluster-bootstrap** intervals that resample original queries (rewrites of one query are never treated as independent; 10,000 resamples, seed 20261017). Tests are two-sided sign-flip permutation tests on the per-query differences, which is a paired randomisation test of the kind compared by Smucker et al. (2007); they are exact when at most 16 differences are non-zero and otherwise use 200,000 random flips.

Holm's step-down correction (Holm, 1979) is applied over the primary family, all 12 system-by-category tests of nDCG@10. Every other p-value, interval and comparison is descriptive. A category whose interval covers zero is reported as "no reliable change detected", never as "no effect".

The comparison of the hybrid with its components (per-query difference in absolute nDCG@10 change) and the Spearman correlation between RBO and the size of the nDCG@10 change over (query, rewrite) pairs are exploratory; the latter has a cluster-bootstrap interval (2,000 resamples).

3.6 Pre-specification, amendments and deviations

The analysis plan (`docs/ANALYSIS_PLAN.md`) was written on 17 October 2026 before any retrieval run on the pilot queries and fixed the systems, categories, measures, estimands, inference and cross-checks. Amendments A1 to A3 (18 October) and the validity rule and reviewer identity (A4 and A5, decided 19 October) preceded the first run on the pilot queries, which started at 2026-10-20T06:02:38+00:00. A4 and A5 were written into the plan file only after the run, so the first 8,130 bytes of the file are exactly the plan as it stood at the run; they hash to `6d31b030288b1c85...`, the value stored in `run_metadata.json`.

The retrieval code had been exercised only on a synthetic collection; the real collection was loaded before the run only to check its counts and to compute the typo-vocabulary diagnostic. Four deviations are recorded in the plan: the executed validity rule is stricter than the plan's text (D1); what the collection cross-check does and does not establish (D2); the blind audit sheet that replaces the older one (D3); and no deviation in systems, measures, estimands, resampling settings, multiplicity family, seeds or depth (D4).

The plan is an internal document, not an external preregistration. It carries no outside timestamp, so the order of events rests on the file hash, the run metadata and, once published, the repository history.

3.7 Verification of the software and the numbers

The package has 85 automated tests. Own nDCG@10, Recall@100 and MRR@10 equal the output of `pytrec_eval` (Van Gysel and de Rijke, 2018) and `ranx` on the same runs (tolerance 10^{-9}); own BM25 scores equal `bm25s`; own RBO equals the `rbo` package and satisfies the analytic identities; the bootstrap interval for a mean agrees with the analytical interval on a large sample and the sign-flip p-value agrees with `scipy.stats.permutation_test` on a small one. A bug-injection check made 23 deliberate one-line breakages in the metric, BM25, stability, fusion, ranking, statistics, analysis, run and rewrite-loading code and the tests caught 22; the survivor is an equivalent mutant (the base of nDCG's logarithm cancels in nDCG/IDCG).

`scripts/verify_results.py` recomputes the effectiveness numbers of this paper from the saved rankings and the judgments with a separate plain implementation and compares them with the tables: every per-query and summary number agrees to machine precision, the intervals and p-values are re-estimated with fresh Monte-Carlo runs under other seeds and agree within sampling error, and the provenance hashes match. The Spearman correlations agree only to within 10^{-4}, because exact ties in the nDCG@10 change are split by floating-point noise in the pipeline's values; resolving the ties by rounding moves the correlations by less than 10^{-4}, which does not affect the two decimals reported.

Every experimental number in this manuscript is filled in from the result files by `scripts/build_paper.py`, which also asserts that each data-dependent sentence below still holds.

The starter script this work began from (a BM25-only scaffold on an Elasticsearch wrapper) was reviewed and not used. It was not run, because it needs an Elasticsearch server. Reading it and the library it calls showed 11 defects, among them an aliased results dictionary that lets later rewrites overwrite the original-query rankings, a stale ranking being scored when a rewrite retrieves nothing, a key that does not exist (so MRR@10 would be silently missing) and a re-index of the whole corpus on every call (`docs/STARTER_CODE_REVIEW.md`). The first three would have produced plausible but wrong numbers in a direction that flatters robustness.

4 Results

4.1 Baselines

On the original queries (Table 2) BM25 reaches nDCG@10 = 0.432 (95% CI 0.375 to 0.489), LSA 0.282 (95% CI 0.232 to 0.334) and the RRF hybrid 0.382 (95% CI 0.326 to 0.437). The BM25 and hybrid intervals overlap substantially, so the pilot does not establish which is better on the original queries (the difference was not tested). LSA has the lowest nDCG@10; its interval is separated from BM25's but overlaps the hybrid's. All values rest on 93 queries and the intervals are wide.

Table 2. Effectiveness on the original queries (93 queries, Vaswani). Mean and 95% percentile bootstrap interval over queries (10,000 resamples).

System	nDCG@10	Recall@100	MRR@10
BM25	0.432 [0.375, 0.489]	0.601 [0.552, 0.649]	0.685 [0.603, 0.763]
LSA	0.282 [0.232, 0.334]	0.495 [0.442, 0.548]	0.506 [0.418, 0.593]
RRF hybrid	0.382 [0.326, 0.437]	0.570 [0.519, 0.621]	0.642 [0.558, 0.723]

4.2 Effect of the variation categories

Table 3 and Figure 1 give the primary results.

Table 3. Effect of query-variation categories on nDCG@10 (primary analysis, 12 tests). Original and rewrite columns are means over the queries that have at least one kept rewrite in the category (each query counts once per category). Relative change is the ratio of the two means minus one; the interval is a cluster bootstrap over original queries. Holm p comes from a two-sided sign-flip permutation test, adjusted over the 12 tests.

System	Category	Queries / rewrites	nDCG@10 original	nDCG@10 rewrites	Relative change, % [95% CI]	Holm p
BM25	misspelling	93 / 279	0.432	0.352	−18.5 [−25.4, −12.1]	<0.001
BM25	paraphrasing	93 / 185	0.432	0.374	−13.3 [−19.7, −6.7]	0.004
BM25	naturality	93 / 172	0.432	0.424	−1.9 [−5.1, 1.1]	1.00
BM25	ordering	8 / 8	0.360	0.360	0.0 [0.0, 0.0]	1.00
LSA	misspelling	93 / 279	0.282	0.222	−21.4 [−29.7, −13.8]	<0.001
LSA	paraphrasing	93 / 185	0.282	0.267	−5.3 [−13.2, 3.7]	1.00
LSA	naturality	93 / 172	0.282	0.281	−0.4 [−3.2, 2.4]	1.00
LSA	ordering	8 / 8	0.312	0.312	0.0 [0.0, 0.0]	1.00
RRF hybrid	misspelling	93 / 279	0.382	0.304	−20.2 [−26.7, −14.2]	<0.001
RRF hybrid	paraphrasing	93 / 185	0.382	0.352	−7.8 [−13.5, −1.7]	0.133
RRF hybrid	naturality	93 / 172	0.382	0.383	0.5 [−1.6, 2.8]	1.00
RRF hybrid	ordering	8 / 8	0.367	0.367	0.0 [0.0, 0.0]	1.00

Misspelling. nDCG@10 changed by −18.5% (95% CI −25.4% to −12.1%) for BM25, −21.4% (95% CI −29.7% to −13.8%) for LSA and −20.2% (95% CI −26.7% to −14.2%) for the hybrid. All three Holm-adjusted p-values are below 0.001 (the permutation test cannot return less than about 5×10^{-6} with 200,000 flips). The losses are of similar size for the three systems and the intervals overlap heavily.

Paraphrasing. BM25 changed by −13.3% (95% CI −19.7% to −6.7%) (Holm p = 0.004). For LSA the change was −5.3% (95% CI −13.2% to 3.7%) (Holm p = 1.00), so no reliable change was detected. For the hybrid it was −7.8% (95% CI −13.5% to −1.7%): the interval excludes zero and the uncorrected permutation

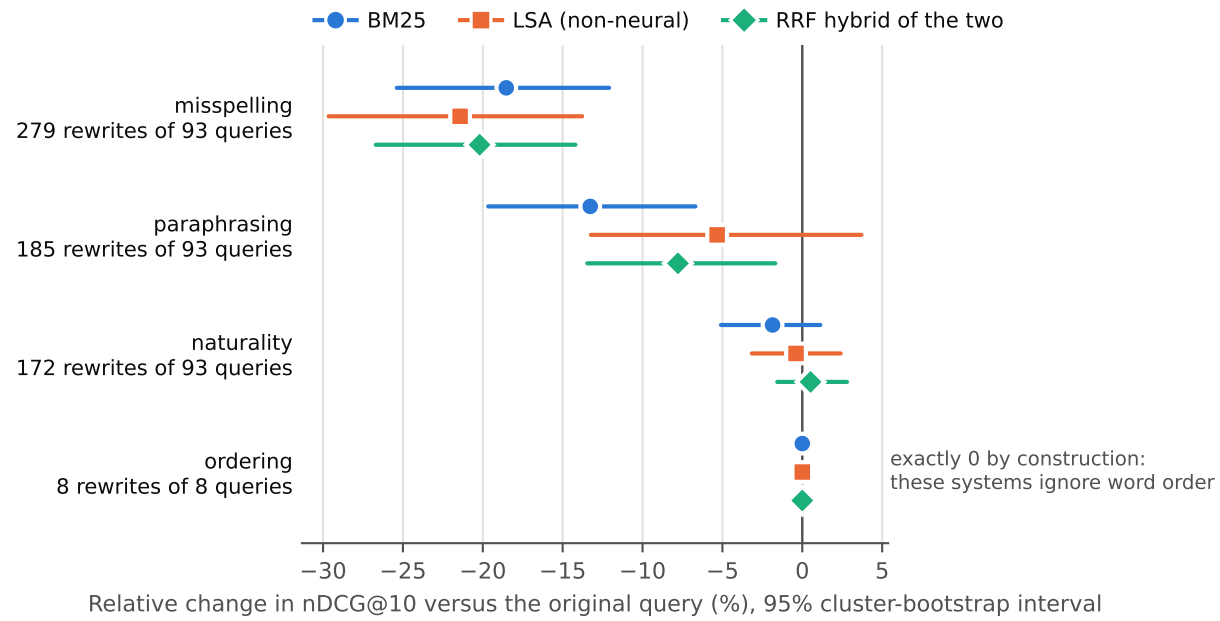


Figure 1. Relative change in nDCG@10 per variation category and system, with 95% cluster-bootstrap intervals (data: Table 3). Ordering is exactly zero because all three systems ignore word order, and rests on only 8 rewrites.

p-value is 0.017, but after Holm correction it is 0.133, which does not meet the pre-specified criterion. Section 4.4 shows that the category average hides a large difference between its two generators.

Naturality. Natural-language questions and stop-word removal changed nDCG@10 by −1.9% (BM25), −0.4% (LSA) and 0.5% (hybrid); all intervals cover zero and no reliable change was detected. The point estimates are below 2% in absolute value.

Ordering. The change is exactly zero for every system, which is guaranteed by construction because BM25, LSA and their fusion are bag-of-words systems. The category has only 8 rewrites. This row is a check that the pipeline is order-invariant, not evidence about word order.

Secondary measures. Recall@100 for misspellings changed by −10.6% (BM25), −12.0% (LSA) and −9.6% (hybrid), and for paraphrasing by −7.1%, −5.5% and −5.6%; all six intervals exclude zero (uncorrected, descriptive). Paraphrasing therefore reduced deep recall for every system even where the nDCG@10 change for LSA was inconclusive. MRR@10 intervals for paraphrasing cover zero for all three systems (BM25 −8.0% (95% CI −16.1% to 1.0%)).

Level versus change. BM25 has the highest nDCG@10 after rewriting in each of misspelling, paraphrasing and naturality (Table 3), followed by the hybrid and then LSA, simply because it starts highest. A system can lose less in relative terms and still be worse in absolute terms.

4.3 Ranking stability and its relation to effectiveness

Table 4 reports how much the top of the ranking moves.

Table 4. Ranking stability: agreement between the ranking for the original query and the ranking for its rewrite, from the same system (1 = identical top 10 / identical ranking; mean over queries, cluster-bootstrap interval).

System	Category	Overlap@10 [95% CI]	RBO, $p = 0.9$ [95% CI]
BM25	misspelling	0.66 [0.62, 0.69]	0.65 [0.62, 0.69]
BM25	paraphrasing	0.55 [0.52, 0.59]	0.53 [0.49, 0.57]
BM25	naturality	0.87 [0.84, 0.90]	0.87 [0.84, 0.90]
BM25	ordering	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
LSA	misspelling	0.69 [0.65, 0.72]	0.66 [0.63, 0.70]
LSA	paraphrasing	0.61 [0.57, 0.65]	0.59 [0.55, 0.63]
LSA	naturality	0.91 [0.89, 0.94]	0.91 [0.88, 0.93]
LSA	ordering	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]
RRF hybrid	misspelling	0.67 [0.63, 0.71]	0.64 [0.60, 0.68]
RRF hybrid	paraphrasing	0.60 [0.56, 0.63]	0.56 [0.52, 0.59]
RRF hybrid	naturality	0.90 [0.88, 0.93]	0.89 [0.86, 0.92]
RRF hybrid	ordering	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]

Misspelling changes the top ten less than paraphrasing does: overlap@10 is 0.66, 0.69 and 0.67 for BM25, LSA and hybrid, against 0.55, 0.61 and 0.60. Yet paraphrasing cost less nDCG@10 than misspelling for every system (Table 3). Across categories, then, ranking stability is not a proxy for harm: a rewrite can reshuffle the top ten into other, equally relevant documents. Across individual (query, rewrite) pairs within a system, however, lower RBO goes with a larger absolute nDCG@10 change: Spearman correlation -0.67 [-0.74, -0.59] for BM25, -0.60 [-0.66, -0.53] for LSA and -0.67 [-0.74, -0.60] for the hybrid (644 pairs; cluster-bootstrap intervals). Part of this association is mechanical, because a ranking that changes more has more room to change its score; the pilot cannot separate that from a substantive link.

4.4 Generators within categories

Table 5. Relative change in nDCG@10 (%) by generator, with 95% cluster-bootstrap intervals. Descriptive only; these 24 cells are not part of the multiplicity-corrected family.

Category	Generator	Rewrites	BM25	LSA	RRF hybrid
misspelling	`NeighbCharSwap`	93	−18.1 [−26.4, −9.8]	−21.3 [−31.2, −11.7]	−20.0 [−28.4, −12.0]
misspelling	`RandomCharSub`	93	−15.5 [−23.7, −7.7]	−17.9 [−27.7, −8.7]	−18.1 [−25.7, −10.8]
misspelling	`QWERTYCharSub`	93	−22.0 [−29.8, −14.3]	−25.0 [−36.0, −14.2]	−22.5 [−31.2, −14.4]
paraphrasing	`LLM_Paraphrase`	93	−8.6 [−15.7, −1.1]	0.9 [−7.9, 10.5]	−0.9 [−7.9, 6.6]
paraphrasing	`LLM_SynonymSwap`	92	−18.5 [−27.6, −9.5]	−12.1 [−23.5, −0.1]	−15.2 [−23.7, −6.6]
naturality	`LLM_NaturalQuestion`	92	−1.7 [−6.6, 3.2]	−1.1 [−5.2, 3.1]	1.1 [−2.4, 4.8]
naturality	`RemoveStopWords`	80	0.1 [−1.4, 1.9]	0.0 [−3.1, 3.1]	−0.1 [−2.1, 2.3]
ordering	`RandomOrderSwap`	8	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Within misspelling all three generators lower nDCG@10 for every system (all intervals below zero), and the QWERTY-neighbour substitution is the most damaging for all three. Within paraphrasing the two generators differ greatly: the synonym swap changes BM25 by -18.5%, about as much as a typo, whereas the structural paraphrase changes it by -8.6%; for LSA the structural paraphrase is indistinguishable from zero (0.9% (95% CI -7.9% to 10.5%)) and the synonym swap costs -12.1%. The synonym swap is the most damaging

non-typo generator for every system. The category average of Table 3 is therefore an average over two different phenomena, and its size depends on how many rewrites each generator contributes. A reading consistent with these data, but not tested here, is that replacing query words by words that do not occur in the relevant documents hurts every system that matches words, while a latent space softens part of that loss. Within naturality both generators are close to zero.

4.5 The hybrid compared with its components (exploratory)

Table 6. Exploratory comparison of the hybrid with its components: mean per-query difference in the *absolute* nDCG@10 change (hybrid minus component) with a 95% cluster-bootstrap interval. Positive means the hybrid lost less (or gained more) than the component. No test, no multiplicity correction.

Category	RRF hybrid minus BM25	RRF hybrid minus LSA
misspelling	+0.003 [−0.014, +0.021]	−0.017 [−0.033, −0.001]
paraphrasing	+0.028 [+0.005, +0.051]	−0.015 [−0.035, +0.005]
naturality	+0.010 [−0.002, +0.024]	+0.003 [−0.004, +0.011]
ordering	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]

In relative terms the hybrid's change lies between those of BM25 and LSA for misspelling and for paraphrasing (Table 3). In the pre-specified, exploratory comparison of absolute changes (hybrid minus component; positive means the hybrid lost less), the difference to BM25 under paraphrasing was +0.028 [+0.005, +0.051], an interval above zero, and there is no clear difference to BM25 under misspelling (+0.003 [−0.014, +0.021]). Against LSA under misspelling the difference was −0.017 [−0.033, −0.001], so the hybrid lost more in absolute terms; this comparison is confounded by level, since LSA starts lower and has less to lose. These contrasts must not be read as "the hybrid is more robust than BM25": after paraphrasing the hybrid's nDCG@10 (0.352) is still below BM25's (0.374), and the hybrid is also below BM25 on the original queries.

4.6 Sensitivity to the validity rule

The whole analysis was repeated with the reviewer's labels alone, without the stricter recheck (47 word-order swaps instead of 8). Every estimate for misspelling and paraphrasing is identical to the main run, because the recheck touches only word-order swaps and stop-word removal. The naturality change moves from −1.9% to −1.8% (BM25), from −0.4% to −0.5% (LSA) and from 0.5% to 0.5% (hybrid), ordering stays exactly zero, no Holm decision changes, and the Spearman correlations change from −0.67, −0.60, −0.67 to −0.70, −0.63, −0.70. The conclusions do not depend on this choice. They do depend on the more basic assumption that the kept rewrites preserve meaning, which this sensitivity analysis cannot test.

4.7 A diagnostic on the typo rewrites

0 of the 279 typo rewrites turned the edited word into another word that occurs in the collection, and the reviewer judged all 30 calibration typos equivalent. For models that match whole words, an edited query word then most likely stops matching altogether, so a typo behaves like deleting the term. This explanation fits the size of the typo losses but was not tested here; the direct test (rewrites that delete the same word) is an obvious next experiment. It need not carry over to retrievers with sub-word tokenisation, which prior work reports are affected by typos too (Zhuang and Zuccon, 2021; Sidiropoulos and Kanoulas, 2022).

5 Discussion

The most robust finding of the pilot is the cost of misspellings: about a fifth of nDCG@10 for all three whole-word systems, with intervals far from zero, similar across systems and of the same order for the three typo generators (Table 5). Losses of this size are in the range that Penha et al. (2022) report for their pipelines (read through a summariser; see Appendix A), but the collections, systems, generators and measures differ, so the numbers are not comparable and no agreement or disagreement is claimed.

The paraphrasing result is more qualified than a first glance at Table 3 suggests. The reliable loss for BM25 comes largely from the synonym swap; the structural paraphrase costs BM25 less and LSA nothing detectable. Whether a category average is informative therefore depends on its generators, and a study that reports only category averages can mislead. This is a caution about reporting, not a new effect. Li et al. (2026) report that LLM-based dense retrievers, although more robust to typos than encoder-only baselines, remain vulnerable to synonym substitution; here the synonym swap is also among the most damaging non-typo variations, but the systems and measures differ and that source was read at abstract level only, so this is a similarity of direction and nothing more.

Stability and harm come apart across categories (Section 4.3). A practitioner who monitors only how much the top ten moves would rank paraphrasing as the more disruptive variation, while the effectiveness loss is larger for misspelling. Both quantities are worth reporting, and neither replaces the other. The within-system correlation also shows that, rewrite by rewrite, instability is informative about harm in this pilot.

For the hybrid, the pilot gives a modest, hedged picture. Its relative losses lie between those of its components, and under paraphrasing it lost less than BM25 in absolute nDCG@10 (exploratory), but its level on the original queries is lower than BM25's. Because both components are whole-word models that share a vocabulary and an analyzer, the hybrid cannot show the complementarity a neural retriever might add. The question that motivated the design, whether a fused lexical-neural first stage is more robust than either component, remains open.

If the typo effect held on other collections, the first thing to check for lexical and LSA-style first stages would be query normalisation or typo-tolerant matching. The pilot cannot say whether it holds elsewhere.

6 Limitations

- **One small, old collection** (93 queries, abstracts from the 1960s), whose provenance and licence are unclear and whose Glasgow page carries a caution. Nothing here generalises to other domains or to modern query logs.
- **No neural retriever.** LSA is non-neural; the hybrid combines two whole-word systems. The pilot says nothing about neural dense retrievers or about neural-plus-lexical hybrids.
- **No human verification of any rewrite.** Validity rests on construction (typos), one LLM reviewer from the generator's model family, and a stricter recheck by the same assistant that wrote the code. No inter-annotator agreement exists. If the kept rewrites are not all meaning-preserving, the effects are overstated.
- **Rewrites written by the same AI system that wrote the code and the analysis.** There was no separate prompt or seed, so the LLM-written rewrites cannot be regenerated.
- **Word order is untestable here**: the three systems ignore it by construction, and only 8 rewrites remain.
- **Unequal contributions.** Rewrites per query differ across generators and categories; the primary analysis averages within query to keep queries independent, but the category averages still depend on

the generator mix (Section 4.4).

- **Multiplicity and power.** Twelve primary tests, 93 queries, wide intervals. The hybrid's paraphrasing result is neither confirmed nor refuted. Generator-level, secondary-measure and hybrid-comparison results are descriptive.
- **Single seed and untuned own BM25.** One LSA seed; BM25 parameters are defaults, not tuned, and the implementation approximates but does not equal Elasticsearch's analyzer.
- **The explanation that typos act like term deletion is a hypothesis.**
- **The literature search was not systematic** and several sources were read only through abstracts or a page summariser (Appendix A); the gap statement may be wrong.
- **AI-authored.** Code, rewrites, review, analysis and text came from an AI system (Section 8). The automated checks reduce but do not remove the chance of error.

7 Reproducibility and availability

The repository contains the package (`src/qvr`), the scripts, the 85 tests, the rewrite files with their review records, the saved rankings of every system for every query and rewrite, the per-query tables, the analysis plan and this manuscript's builder. The run took about 10.02 seconds on one CPU in the sandbox (Python 3.13.16, numpy 2.5.4, scipy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, snowballstemmer 3.1.1). The commands are `python scripts/fetch_vaswani.py`, `python scripts/run_experiment.py ...`, `python scripts/analyze.py ...`, `python scripts/build_paper.py` and `python scripts/verify_results.py`; the README lists them in full. The BM25, LSA and RRF code is deterministic given the seeds; a re-run in a fresh environment with other library versions gave the same per-query tables and the same document order in every ranked list, with stored LSA scores differing only in the seventh decimal (`results/reproduction_check.md`). The repository address is [REPOSITORY URL: to be confirmed by the author]. The Vaswani files are not included; `fetch_vaswani.py` retrieves and checks them.

8 AI assistance, authorship and responsibility

[HUMAN AUTHOR TO COMPLETE AND CONFIRM. This section was drafted by the AI system and does not replace the authors' own statement.]

The software, the tests, the 279 LLM-written query rewrites, the experiment runs, the analysis plan, the checks and this text were produced in one working session with Claude (Anthropic); the session was configured as Claude Sonnet 5.5, and the serving model cannot be verified from inside the session. Code produced 455 further rewrite candidates. A smaller Claude model setting acted as the blind reviewer of rewrite validity. The AI system cannot be an author and cannot take responsibility for the content; arXiv's policy states that generative AI tools should not be listed as authors, that authors take full responsibility for all contents irrespective of how they were generated, and that significant use of such tools should be reported (Boboris, 2023). Before this work is submitted anywhere, the human author(s) must at least:

- read the whole manuscript, the code and the data, and decide what is claimed;
- re-run `pytest` and `python scripts/verify_results.py` and confirm the numbers;
- check every citation against the cited paper, in particular the claims that rest on abstracts or on a page summariser (Appendix A);
- decide whether a pilot of this scope should be submitted at all, or first extended (Appendix B);
- write their own statement of contributions and of AI use that matches what they actually did.

9 Conclusion

On one small collection, misspellings cost BM25, LSA and their RRF hybrid about a fifth of nDCG@10 (all Holm-adjusted p-values below 0.001). Paraphrasing cost BM25 13.3% but did not reliably change LSA or, after correction, the hybrid. Natural-question and stop-word rewrites changed little, and word order cannot matter for these systems. Ranking stability and effectiveness loss are linked within a system but not comparable across categories. The pilot's value is a verified, reproducible pipeline and a set of effect sizes; its weaknesses are the absence of neural retrievers, of other collections and of any human validation of the rewrites. The research question that motivated it, how a fused lexical-neural first stage behaves under query variation, is not answered.

References

Marwah Alaofi, Luke Gallagher, Mark Sanderson, Falk Scholer, and Paul Thomas. Can generative LLMs create query variants for test collections? an exploratory study. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1869–1873. ACM, 2023. doi: 10.1145/3539618.3591960.

Peter Bailey, Alistair Moffat, Falk Scholer, and Paul Thomas. UQV100: A test collection with query variability. In *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 725–728. ACM, 2016. doi: 10.1145/2911451.2914671.

Peter Bailey, Alistair Moffat, Falk Scholer, and Paul Thomas. Retrieval consistency in the presence of query variations. In *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 395–404. ACM, 2017. doi: 10.1145/3077136.3080839.

Kat Boboris. arXiv announces new policy on ChatGPT and similar tools. arXiv blog, 31 January 2023, https://blog.arxiv.org/2023/01/31/arxiv-announces-new-policy-on-chatgpt-and-similar-tools, 2023. Accessed 22 October 2026.

Stefano Campese, Alessandro Moschitti, and Ivano Lauriola. Improving document retrieval coherence for semantically equivalent queries. arXiv preprint arXiv:2508.07975, 2025.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. ACM, 2009. doi: 10.1145/1571941.1572114.

Scott Deerwester, Susan T. Dumais, George W. Furnas, Thomas K. Landauer, and Richard Harshman. Indexing by latent semantic analysis. *Journal of the American Society for Information Science*, 41(6): 391–407, 1990.

Tim Hagen, Harrisen Scells, and Martin Potthast. Revisiting query variation robustness of transformer models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4283–4296, Miami, Florida, USA, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.248.

S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2): 65–70, 1979.

ir_datasets. Vaswani. https://ir-datasets.com/vaswani.html, 2026. Accessed 20 October 2026.

Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002. doi: 10.1145/582415.582418.

Yongkang Li, Panagiotis Eustratiadis, Yixing Fan, and Evangelos Kanoulas. On the robustness of LLM-based dense retrievers: A systematic analysis of generalizability and stability. arXiv preprint arXiv:2604.16576, 2026. Version 1, 17 April 2026; not peer reviewed as far as the arXiv page shows.

Alistair Moffat, Falk Scholer, Paul Thomas, and Peter Bailey. Pooled evaluation over query variations. In *Proceedings of the 24th ACM International Conference on Information and Knowledge Management*, pages 1759–1762. ACM, 2015. doi: 10.1145/2806416.2806606.

Gustavo Penha, Arthur Câmara, and Claudia Hauff. Evaluating the robustness of retrieval pipelines with query variation generators. In *Advances in Information Retrieval (ECIR 2022)*, Lecture Notes in Computer Science, pages 397–412. Springer, 2022. doi: 10.1007/978-3-030-99736-6_27. arXiv:2111.13057.

Sezen Perçin, Xin Su, Qutub Sha Syed, Phillip Howard, Aleksei Kuvshinov, Leo Schwinn, and Kay-Ulrich Scholl. Investigating the robustness of retrieval-augmented generation at the query level. arXiv preprint arXiv:2507.06956, 2025. Version 1, 9 July 2025.

M. F. Porter. An algorithm for suffix stripping. *Program: Electronic Library and Information Systems*, 14(3): 130–137, 1980. doi: 10.1108/eb046814.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019.

Georgios Sidiropoulos and Evangelos Kanoulas. Analysing the robustness of dual encoders for dense retrieval against misspellings. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2132–2136. ACM, 2022. doi: 10.1145/3477495.3531818.

Georgios Sidiropoulos and Evangelos Kanoulas. Improving the robustness of dense retrievers against typos via multi-positive contrastive learning. arXiv preprint arXiv:2403.10939, 2024. Version 1, 16 March 2024.

Mark D. Smucker, James Allan, and Ben Carterette. A comparison of statistical significance tests for information retrieval evaluation. In *Proceedings of the 16th ACM Conference on Information and Knowledge Management*, pages 623–632. ACM, 2007. doi: 10.1145/1321440.1321528.

Panuthep Tasawong, Wuttikorn Ponwitayarat, Peerat Limkonchotiwat, Can Udomcharoenchaikit, Ekapol Chuangsuwanich, and Sarana Nutanong. Typo-robust representation learning for dense retrieval. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 1106–1115, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-short.95.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogenous benchmark for zero-shot evaluation of information retrieval models. NeurIPS 2021 Datasets and Benchmarks Track; arXiv:2104.08663, 2021.

University of Glasgow, Information Retrieval Group. NPL test collection (resources page). http://ir.dcs.gla.ac.uk/resources/test_collections/npl/, 2026. Accessed October 2026.

Christophe Van Gysel and Maarten de Rijke. Pytrec_eval: An extremely fast python interface to trec_eval. In *Proceedings of the 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 873–876. ACM, 2018. doi: 10.1145/3209978.3210065.

William Webber, Alistair Moffat, and Justin Zobel. A similarity measure for indefinite rankings. *ACM Transactions on Information Systems*, 28(4), 2010. doi: 10.1145/1852102.1852106.

Shengyao Zhuang and Guido Zuccon. Dealing with typos for BERT-based passage retrieval and ranking. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 2836–2842. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.emnlp-main.225.

Shengyao Zhuang and Guido Zuccon. CharacterBERT and self-teaching for improving the robustness of dense retrievers on queries with typos. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1444–1454. ACM, 2022. doi: 10.1145/3477495.3531951.

Appendix A: Literature search record

What was done. A general web search engine (standard and extended modes; results are titles and links only) and direct reads of arXiv listings, the ACL Anthology, Crossref records and publisher pages. DBLP could not be used (blocked for automated access). Google Scholar, the ACM Digital Library and Semantic Scholar search were not used. No systematic-review protocol was followed. Inclusion rule applied by title and abstract: the item studies the robustness of retrieval (first stage, dense or re-ranking) to the wording of queries (typos, paraphrases, variants) or the consistency of rankings across variants. Items about retrieval-augmented generation only, generative retrieval only, or adversarial corpus attacks were noted but not used. The record of searches made at the very start of the work was lost when the working session's context was compacted; the searches below are those that could be recovered from session logs or were made afterwards. A human author should repeat and log the search before relying on the gap statement.

Table 7. Searches that could be recovered or were made afterwards (dates are those of the run; the earlier ones are from the sub-agent's log, in UTC).

#	Date	Query	Mode
1	2026-10-17	query variation robustness first-stage retrieval BM25 dense hybrid	standard
2	2026-10-17	typos robustness hybrid retrieval BM25 dense fusion	standard
3	2026-10-17	paraphrase robustness BEIR dense retrievers query reformulation sensitivity	standard
4	2026-10-17	ranking stability query variants rank-biased overlap neural retrieval	standard
5	2026-10-17	LLM-generated query variants meaning-preserving validation retrieval robustness benchmark	extended
6	2026-10-17	query variations robustness retrieval 2026 arXiv	extended
7	2026-10-17	"query variation" retrieval robustness	standard
8	2026-10-17	typos dense retrieval	standard
9	2026-10-17	paraphrase robustness retrieval	standard
10	2026-10-17	hybrid retrieval robustness	standard
11	2026-10-22	hybrid retrieval BM25 dense reciprocal rank fusion robustness to query typos and paraphrases evaluation	standard
12	2026-10-22	query variation generators misspelling ordering naturality paraphrasing retrieval robustness first-stage retrievers	standard
13	2026-10-22	lexical versus dense retrieval typo robustness BM25 misspelled queries comparison study	standard
14	2026-10-22	ranking stability across query variants rank-biased overlap relationship with effectiveness change	standard
15	2026-10-22	LLM-generated query variants validity human judgment same information need retrieval evaluation	standard

Searches 11 and 12 again returned the papers already known (Penha et al., 2022; Li et al., 2026) and no study of a fused lexical-semantic first stage under query variations; search 10 returned nothing on that topic either. Searches 13 to 15 returned the typo-robustness, consistency and LLM-query-variant papers already cited, plus the items listed below as seen by title only. The results are only titles, so "returned nothing" means that no title suggested it.

How each cited source was read. "Record" means a bibliographic record only (title, authors, venue, pages). "Abstract" means the abstract or listing page. "Summariser" means the full text through a page-summarising tool, whose output can be wrong and must be checked against the PDF.

Table 8. Reading level of each cited source.

Source	Read	Used for
Penha et al. (2022)	abstract page, Crossref record, full text through a summariser, authors' repository README	taxonomy, generators, datasets and systems used, size of the average loss, validation by the authors
Hagen et al. (2024)	ACL Anthology page and abstract, Crossref record, full text through a summariser	scope (re-rankers, newer models), statement that the re-ranking stage is evaluated
Li et al. (2026)	arXiv abstract page	robustness of LLM-based dense retrievers, typo and synonym findings
Perçin et al. (2025)	arXiv abstract page	RAG retrievers degrade under minor variations
Campese et al. (2025)	arXiv page, abstract summary only (submission date not shown)	consistency loss for equivalent queries
Sidiropoulos and Kanoulas (2022), Sidiropoulos and Kanoulas (2024)	Crossref record; arXiv abstract page	typo robustness of dense retrievers
Zhuang and Zuccon (2021), Tasawong et al. (2023)	ACL Anthology pages	typo robustness of neural retrieval
Zhuang and Zuccon (2022)	Crossref record	typo robustness of dense retrievers
Alaofi et al. (2023)	Crossref record only	LLM-generated query variants
Bailey et al. (2016), Bailey et al. (2017), Moffat et al. (2015)	bibliographic records	query variability and consistency
Thakur et al. (2021)	arXiv listing (version 4); venue taken from a proceedings link in search results, not opened	BEIR, planned follow-up
Cormack et al. (2009), Webber et al. (2010), Järvelin and Kekäläinen (2002), Smucker et al. (2007), Van Gysel and de Rijke (2018)	bibliographic records (Crossref, Webis anthology)	methods used; implementations checked against independent software
Robertson and Zaragoza (2009), Deerwester et al. (1990), Porter (1980)	publisher or index pages	methods used
Holm (1979)	GESIS knowledge-graph record (title, journal, volume, issue, pages; the author is inferred from the record identifier)	multiplicity correction
ir_datasets (2026), University of Glasgow, Information Retrieval Group (2026)	the web pages	collection description
Boboris (2023)	the arXiv blog post	AI-use and authorship policy

Items seen in search results by title only, not read and not relied on: on robustness of generative retrieval (arXiv 2306.12756; arXiv 2412.18768), adaptive RAG robustness to query variations (arXiv 2604.10745), a study of LLM-based retrievers on BRIGHT including query perturbations (arXiv 2604.03676), query rewriting for retrieval (arXiv 2407.12529, 2603.13301), zero-shot hybrid retrieval models (ECIR 2022), demographically inspired LLM query variants (arXiv 2508.17644), LLM query simulation and answer-side intrusion (arXiv 2608.25245), and rank-similarity measures based on maximised effectiveness difference (arXiv 1408.3587).

Claims that depend on a summariser's reading of full text and must be checked by the human author: the roughly 20% average loss and the datasets and systems used by Penha et al. (2022); that neither Penha et al. (2022) nor Hagen et al. (2024) evaluates a fused lexical-semantic first stage; that Hagen et al. (2024) evaluate the re-ranking stage.

Appendix B: What a stronger study needs

1. At least two neural dense retrievers, chosen and documented before running, with their hybrids with BM25, on at least three BEIR domains (instructions: `docs/RUN_BEIR_LOCALLY.md`).
2. Human validation of the rewrites by at least two annotators, with agreement reported (the blind audit sheet and tool are in the repository), and logged prompts and decoding settings for the LLM-written rewrites.
3. A term-deletion control for the typo rewrites and a generator-level pre-specification for paraphrasing.
4. A systematic literature search with a written protocol, a log of queries and databases, and a full-text reading of the closest papers.
5. A pre-registered analysis plan with an external timestamp.